RailMitra - AI-enabled railway monitoring platform

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Problem Statement :

Railway safety is being damaged not just by the defects themselves, but by the gap between the actual condition of assets and the maintenance action taken. In many cases, the current railway maintenance still is reactive, schedule-based, or limited to rule-based diagnostics, i.e., interventions may be too late for safety or too early for cost-efficiency.
Audit and research evidence supports this framing. According to a CAG audit report on Indian Railways in 2022, the primary reasons for derailments were track maintenance, deviation of track parameters, wrong setting of points, and defects in rolling stock. The report also noted shortfalls in inspections and mentioned that out of 1,127 derailments between 2017 and 2021, 289 were due to track renewals. Research surveys also point out that maintenance is resource-heavy, increasingly complex and still limited by fragmented data, weak interoperability and low-quality monitoring pipelines.

Railway infrastructure faces constant stress due to load, vibration, weather exposure, and long operational hours. Small defects often grow slowly and may not be visible during routine manual inspection, which creates a risk of delayed maintenance and possible failure.

Existing Problems

1. Heavy dependence on manual or periodic inspection instead of continuous monitoring.
2. Delayed identification of vibration abnormalities, overheating, and track irregularities.
3. Reactive maintenance increases downtime and emergency repair effort.
4. Maintenance teams may not get clear risk-based prioritization for which segment needs attention first.
5. Large railway networks generate too many monitoring points for only manual supervision.

The problem is not the lack of maintenance activity, it is the lack of maintenance intelligence

Existing solutions

Today's solutions broadly fall into four buckets:

Existing approach	What it does	Where it fails
Reactive maintenance	Fix after failure or visible fault.	Too late for safety-critical assets; causes unplanned downtime and higher breakdown cost.
Preventive maintenance	Inspect/replace at fixed intervals.	Causes over-maintenance or missed early degradation because calendar schedule is not real condition.
Condition-based monitoring	Uses inspections, geometry checks, vibration, temperature, images, event logs.	Usually subsystem-specific and not fully integrated into one decision engine.

There are Existing solutions, but they are mostly partial. Inspections, ultrasonic flaw detection, track recording, wheel monitoring and some condition monitoring systems are already in use in the railways. Indian Railways has also stressed on web enabled monitoring and better maintenance traceability. The real gap is that these systems tend to be siloed, periodic, staff-dependent, and weak at transforming raw detection into prioritized action

Proposed Solution

RailMitra is proposed as an AI-enabled railway monitoring platform that converts sensor signals into useful maintenance decisions. It supports data-driven inspection and can be used as a smart decision-support tool for railway operators.

What RailMitra Does

1. Collects track or rolling sensor values such as vibration, acceleration, and temperature.

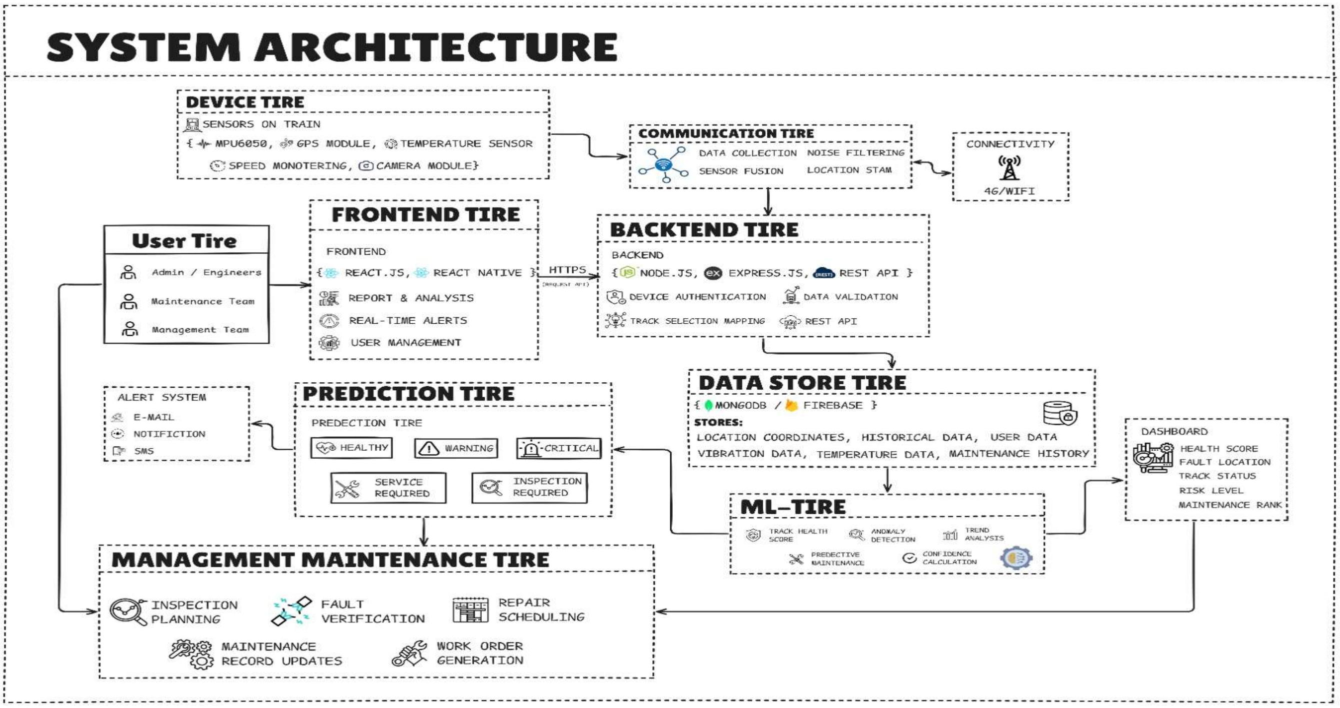
2. Processes the incoming data and derives useful indicators for model input.

3. Uses machine learning to classify the condition of a track segment as Normal, Inspect, or Urgent.

4. Generates alerts for risky segments and displays them on a dashboard for quick review.

5. Helps maintenance teams take action before breakdown or service disruption happens.

System Architectue



Technology Stack

Layer	Technology	Purpose
Frontend	React.js	Interactive dashboard UI
Styling	CSS / component styling	Responsive visualization
Backend	Node.js, Express.js	APIs and business logic
Database	MongoDB	Store sensor data, predictions, alerts
ML Layer	Python / ML model	Risk prediction and classification
Data Source	Sensors / simulated sensor feed	Input for monitoring

Working Methodology

1. RailMitra works in a simple pipeline from sensing to action. The methodology below explains the system flow clearly.
2. Sensors capture values such as vibration, acceleration, and temperature from a track segment or related railway component.
3. The backend receives the readings through APIs or simulated live input.
4. Raw data is cleaned, formatted, and prepared for model processing.
5. The ML model extracts useful patterns and predicts the condition class and risk score.
6. Prediction results are stored in the database for analysis and history tracking.
7. The dashboard displays current values, alerts, trend charts, and segment-level results
8. Maintenance teams use the output to inspect, review, or close alerts based on severity.

Innovation & Uniqueness

This section is the strongest value proposition of RailMitra.

RailMitra is not just a dashboard and not just an ML model. It combines sensing, prediction, visualization, and action workflow in one practical railway-focused system. Railway predictive maintenance is important because early warning from sensor data can prevent failures before they become dangerous or costly.

Why RailMitra is Innovative

1) It shifts railway maintenance from fixed schedule checking to condition-based intelligence.
2) It converts raw sensor signals into clear classes such as Normal, Inspect, and Urgent, which is easier for operational teams to use.
3) It supports near real-time decision-making instead of only offline reporting.
4) It is scalable because new sensors, routes, and models can be added later.
5) It creates a direct bridge between AI prediction and maintenance action through alerts and dashboard review.

Impact

Social Impact
A safer railway network benefits passengers, staff, and nearby communities. Early detection of risky segments can reduce the chance of accidents caused by hidden infrastructure problems.

Economic Impact
Predictive maintenance helps reduce emergency repair cost, lowers unplanned downtime, and improves asset life through better maintenance timing. Better prioritization also helps optimize manpower and spare part usage.

Operational Impact
RailMitra can improve maintenance planning, make inspections more focused, and increase visibility across many track segments. This supports more reliable rail operations and faster response to emerging issues.

Challenges & Limitations

RailMitra is a strong prototype, but some practical challenges still exist.

Real sensor integration may require hardware setup, calibration, and field testing.

Model quality depends on the quantity and quality of training data.

False positives and false negatives can affect maintenance decisions.

Railway environments are complex, so a production system would need more validation under different operating conditions.

Live deployment may also need stronger data security, device management, and large-scale infrastructure support.

Future Scope

RailMitra can be expanded significantly beyond the current hackathon prototype.

Integration with real IoT sensor networks.

Use of advanced deep learning and anomaly detection models.

GIS or map-based visualization of risky railway segments.

Mobile app support for field inspection teams.

Automatic maintenance scheduling recommendations.

Computer vision support using cameras for visible defect detection.

Larger multi-route and multi-zone deployment with centralized monitoring.

Business Model

RailMitra can be developed as a B2B railway technology solution for railway authorities, metro systems, industrial rail corridors, and private operators.

Possible Model
Software-as-a-Service subscription for dashboard and analytic.
Hardware plus software bundle for sensor-based deployment.
Enterprise licensing for railway departments.
Annual maintenance and support contracts.
Paid customization for region-specific monitoring needs.

Business Value
The business value comes from reducing downtime, improving safety, and helping operators take earlier action through predictive intelligence. Solutions that improve visibility and lower maintenance uncertainty are attractive because they create measurable operational benefit.

Conclusion

RailMitra is a practical hackathon solution for smart railway predictive maintenance. It addresses a real problem by combining sensors, machine learning, backend intelligence, and a dashboard into one simple system that helps detect risk early and support better maintenance decisions.

The project is innovative because it turns raw condition data into usable action. With further real-world integration and model improvement, RailMitra has the potential to become a scalable safety and operations platform for modern railways.

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